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Lossless compression for low-latency video transmission in resource-constrained encoding environment

Abstract

This disclosure provides methods, devices, and systems for data compression. The present implementations more specifically relate to lossless data compression techniques that combine entropy coding with run-length encoding of zeroes. In some aspects, an encoder may obtain a sequence of integer values representing a frame of data and may iteratively scan the sequence for runs of zeros. During each iteration of the scan, the encoder may increment a count value (N) if the current integer value is equal to zero or, if the current integer value is not equal to zero, output a pattern of bits representing the current count value or the current integer value. In some implementations, the pattern of bits may include a codeword associated with an entropy encoding scheme. More specifically, the encoder may encode the current integer value as the codeword if $N > 0$ and may encode the current count value as the codeword if $N = 0$.

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Background/Summary

TECHNICAL FIELD

(1) The present implementations relate generally to data compression, and specifically to lossless

compression for low-latency video transmissions in resource-constrained encoding environments.

BACKGROUND OF RELATED ART

(2) Data compression is a technique for encoding information into (generally) smaller units of data. As such, data compression can be used to reduce the bandwidth or overhead needed to store or transmit such information over a communications channel (such as a wired or wireless medium). For example, an encoder encodes or compresses the information into a sequence of coded bits (also referred to as a “codeword”) and a decoder subsequently decodes or decompresses the codeword to recover the original information. Data compression techniques can be generally categorized as “lossy” or “lossless.” Lossy data compression may result in some loss of information between the encoding and decoding of such information. In contrast, no information is lost as a result of encoding or decoding such information using lossless data compression.

(3) Entropy encoding is a form of lossless data compression that encodes data values into codewords of varying lengths based on the probability of occurrence of each data value. For example, data values that have a higher probability of occurrence may be encoded into shorter codewords than data values that have a lower probability of occurrence. As such, entropy encoding is often used for image compression in video playback applications (where adjacent pixels, in space or time, are likely to have the same or similar values). However, existing entropy encoding techniques (such as Huffman coding and arithmetic coding) rely on complex arithmetic or high memory bandwidth. Thus, new data compression techniques may be needed to meet the low latency requirements of modern video playback applications in resource-constrained environments.

SUMMARY

(4) This Summary is provided to introduce in a simplified form a selection of concepts that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to limit the scope of the claimed subject matter.

(5) One innovative aspect of the subject matter of this disclosure can be implemented in a method performed by an encoder. The method includes steps of obtaining a number (M) of integer values representing a frame of data; performing a zero-detection operation that scans the M integer values over M iterations, respectively, and indicates, during each of the M iterations, whether the scanned integer value is equal to zero; incrementing a count value (N) responsive to the zero-detection operation indicating that the scanned integer value is equal to zero; and outputting a first pattern of bits responsive to the zero-detection operation indicating that the scanned integer value is not equal to zero, the first pattern of bits representing one of the count value or the scanned integer value based on whether N is greater than zero.

(6) Another innovative aspect of the subject matter of this disclosure can be implemented in an encoder that includes a processing system and a memory. The memory stores instructions that, when executed by the processing system, cause the encoder to obtain a number (M) of integer values representing a frame of data; perform a zero-detection operation that scans the M integer values over M iterations, respectively, and indicates, during each of the M iterations, whether the scanned integer value is equal to zero; increment a count value (N) responsive to the zero-detection operation indicating that the scanned integer value is equal to zero; and output a first pattern of bits responsive to the zero-detection operation indicating that the scanned integer value in the sequence is not equal to zero, the first pattern of bits representing one of the count value or the scanned integer value based on whether N is greater than zero.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present implementations are illustrated by way of example and are not intended to be

limited by the figures of the accompanying drawings.

(2) FIG. 1 shows an example communication system for encoding and decoding image data.

(3) FIG. 2 shows a block diagram of a video transmission system, according to some implementations.

(4) FIG. 3 shows a block diagram of a variable-length encoder, according to some implementations.

(5) FIG. 4 shows an example zero-run compression encoder, according to some implementations.

(6) FIG. 5 shows an example encoding of differential image data, according to some implementations.

(7) FIG. 6 shows a block diagram of an encoder, according to some implementations.

(8) FIG. 7 shows an illustrative flowchart depicting an example operation for data encoding, according to some implementations.

DETAILED DESCRIPTION

(9) In the following description, numerous specific details are set forth such as examples of specific components, circuits, and processes to provide a thorough understanding of the present disclosure. The term “coupled” as used herein means connected directly to or connected through one or more intervening components or circuits. The terms “electronic system” and “electronic device” may be used interchangeably to refer to any system capable of electronically processing information. Also, in the following description and for purposes of explanation, specific nomenclature is set forth to provide a thorough understanding of the aspects of the disclosure. However, it will be apparent to one skilled in the art that these specific details may not be required to practice the example embodiments. In other instances, well-known circuits and devices are shown in block diagram form to avoid obscuring the present disclosure. Some portions of the detailed descriptions which follow are presented in terms of procedures, logic blocks, processing and other symbolic representations of operations on data bits within a computer memory.

(10) These descriptions and representations are the means used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. In the present disclosure, a procedure, logic block, process, or the like, is conceived to be a self-consistent sequence of steps or instructions leading to a desired result. The steps are those requiring physical manipulations of physical quantities. Usually, although not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated in a computer system. It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities.

(11) Unless specifically stated otherwise as apparent from the following discussions, it is appreciated that throughout the present application, discussions utilizing the terms such as “accessing,” “receiving,” “sending,” “using,” “selecting,” “determining,” “normalizing,” “multiplying,” “averaging,” “monitoring,” “comparing,” “applying,” “updating,” “measuring,” “deriving” or the like, refer to the actions and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

(12) In the figures, a single block may be described as performing a function or functions; however, in actual practice, the function or functions performed by that block may be performed in a single component or across multiple components, and/or may be performed using hardware, using software, or using a combination of hardware and software. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described below generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the

described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure. Also, the example input devices may include components other than those shown, including well-known components such as a processor, memory and the like.

(13) The techniques described herein may be implemented in hardware, software, firmware, or any combination thereof, unless specifically described as being implemented in a specific manner. Any features described as modules or components may also be implemented together in an integrated logic device or separately as discrete but interoperable logic devices. If implemented in software, the techniques may be realized at least in part by a non-transitory processor-readable storage medium including instructions that, when executed, performs one or more of the methods described above. The non-transitory processor-readable data storage medium may form part of a computer program product, which may include packaging materials.

(14) The non-transitory processor-readable storage medium may comprise random access memory (RAM) such as synchronous dynamic random-access memory (SDRAM), read only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, other known storage media, and the like. The techniques additionally, or alternatively, may be realized at least in part by a processor-readable communication medium that carries or communicates code in the form of instructions or data structures and that can be accessed, read, and/or executed by a computer or other processor.

(15) The various illustrative logical blocks, modules, circuits and instructions described in connection with the embodiments disclosed herein may be executed by one or more processors (or a processing system). The term “processor,” as used herein may refer to any general-purpose processor, special-purpose processor, conventional processor, controller, microcontroller, and/or state machine capable of executing scripts or instructions of one or more software programs stored in memory.

(16) Various aspects relate generally to data compression, and more particularly, to lossless data compression techniques that combine entropy coding with run-length encoding of zeroes. In some aspects, an encoder may obtain a sequence of integer values representing a frame of data and may iteratively scan the sequence for runs of zeros. During each iteration of the scan, the encoder may increment a count value (N) if the current integer value is equal to zero or, if the current integer value is not equal to zero, output a pattern of bits representing the current count value or the current integer value. In some implementations, the pattern of bits may include a codeword associated with an entropy encoding scheme (such as exponential Golomb-Rice). More specifically, the encoder may encode the current integer value as the codeword if $N > 0$ and may encode the current count value as the codeword if $N = 0$. In some implementations, the encoder may append a zero-run (ZR) prefix to the beginning of the codeword if $N = 0$. For example, the ZR prefix may signal that the following codeword represents a run of zeroes (rather than a symbol of data). In some implementations, the ZR prefix may be a single bit having a different binary value than the least-significant bit of the codeword.

(17) Particular implementations of the subject matter described in this disclosure can be implemented to realize one or more of the following potential advantages. The data compression techniques of the present disclosure may improve the latency of video transmissions, particularly in resource-constrained environments. As described above, adjacent pixels in video frames are likely to have the same or similar values. As such, many video encoders utilize differential encoding to transmit only the differences in pixel values with each frame of data. Such differential data values are likely to be relatively small, with zero more likely to occur than any other value. Thus, by combining entropy encoding of non-zero values with run-length encoding of zeroes, aspects of the present disclosure may achieve high compression ratios for video transmissions with relatively low encoding complexity. More specifically, by adding a ZR prefix to a given codeword, aspects of the present disclosure may leverage inherent properties of existing entropy encoding techniques to

signal that the codeword represents a run of zeroes. For example, all exponential Golomb-Rice codewords include a unary prefix consisting exclusively of zeroes (or ones). As such, inserting a “1” bit before the unary prefix creates an impossible codeword, which can be interpreted by a decoder as a number of consecutive zeroes.

(18) FIG. 1 shows an example communication system **100** for encoding and decoding image data. The communication system **100** includes an encoder **110** and a decoder **120**. The encoder **110** and decoder **120** may be provided in respective communication devices such as, for example, computers, switches, routers, hubs, gateways, cameras, displays, or other devices capable of transmitting or receiving communication signals.

(19) The encoder **110** receives input data **102** to be transmitted or stored via a channel **130**. For example, the channel **130** may include a wired or wireless transmission medium that facilitates communications between the encoder **110** and the decoder **120**. Alternatively, or in addition, the channel **130** may include a data storage medium. In some aspects, the encoder **110** may be configured to compress the size of the input data **102** to accommodate the bandwidth, storage, or other resource limitations associated with the channel **130**. For example, the encoder **110** may encode each unit of input data **102** as a respective “codeword” that can be transmitted or stored over the channel **130** (as encoded data **104**). The decoder **120** is configured to receive the encoded data **104** via the channel **130** and decode the encoded data **104** as output data **106**. For example, the decoder **120** may decompress or otherwise reverse the compression performed by the encoder **110** so that the output data **106** is substantially similar, if not identical, to the original input data **102**.

(20) Data compression techniques can be generally categorized as “lossy” or “lossless.” Lossy data compression may result in some loss of information between the encoding and decoding steps. As such, the output data **106** may be different than the input data **102**. In contrast, lossless data compression does not result in any loss of information between the encoding and decoding steps as long as the channel **130** does not introduce errors into the encoded data **104**. As such, the output data **106** is identical to the input data **102**. Entropy encoding is a form of lossless data compression that encodes data values into codewords of varying lengths based on the probability of occurrence of each data value. For example, data values that have a higher probability of occurrence may be encoded into shorter codewords than data values that have a lower probability of occurrence. Example entropy encoding techniques including, among other examples, arithmetic coding, Huffman coding, and Golomb coding.

(21) Exponential Golomb-Rice (EGR) is a particular type of Golomb coding that encodes integer values into codewords of varying lengths based, at least in part, on the number (n) of bits needed to represent each integer value as a binary number. More specifically, each EGR codeword consists of a unary prefix, a delimiter, and a remainder. The unary prefix represents the least-significant bits (LSBs) of the EGR codeword and consists of n consecutive 0's (or 1's), the delimiter is a bit having a binary value equal to 1 (or 0), and the remainder represents the most-significant bits (MSBs) of the EGR codeword and consists of the $n-1$ LSBs of the binary number. For example, the binary number “100” (corresponding to the integer value “4”) can be encoded as the EGR codeword “001000,” where the unary prefix (corresponding to the three 0's to the right of the “1” bit) indicates that there are three bits in the binary number “100” and the remainder (corresponding to the two 0's to the left of the “1” bit) represents the two LSBs of the binary number “100.”

(22) Because the length of an EGR codeword depends on the number of bits associated with the binary number to be encoded, larger integer values are encoded as longer EGR codewords whereas smaller integer values are encoded as shorter EGR codewords. As such, EGR coding is well-suited for image compression in video playback applications. For example, a video source may transmit only the changes or differences in pixel values associated with successive frames of image data to be displayed or played back on a display device. This technique is referred to more generally as differential encoding. Because adjacent pixels (in space or time) are likely to have the same or similar pixel values, differential encoding of image data results in a relatively normal (or Gaussian)

distribution of data values centered around zero. In other words, smaller data values (closer to zero) have a higher probability of occurring than larger data values. Thus, EGR coding may achieve a high compression ratio when used to encode differential image data.

(23) FIG. 2 shows a block diagram of a video transmission system **200**, according to some implementations. The video transmission system **200** includes an image encoder **210** and an image decoder **220**. In some implementations, the image encoder **210** and image decoder **220** may be examples of the encoder **110** and decoder **120**, respectively, of FIG. 1. Thus, the image encoder **210** may be communicatively coupled to the image decoder **220** via a channel (such as the channel **130** of FIG. 1).

(24) The image encoder **210** is configured to encode raw image data **201** for transmission, as compressed image data **203**, to the decoder **220**. For example, the raw image data **201** may include an array of pixels (or pixel values) representing a digital image captured or acquired by an image source (such as a camera or other image output device). The image decoder **220** is configured to decode the compressed image data **203** for display or playback, as reconstructed image data **205**, on a display device (such as a television, computer monitor, smartphone, or any other device that includes an electronic display). More specifically, the image decoder **220** is configured to reverse the encoding performed by the image encoder **210** so that the reconstructed image data **205** is substantially similar, if not identical, to the raw image data **201**. In some implementations, the image encoder **210** may be configured to transmit to the image decoder **220** a sequence of frames of compressed image data **203** each representing a respective image or frame of video.

(25) In some implementations, the image encoder **210** may include a differential encoding component **212** and a data compression component **214**. The differential encoding component **212** is configured to encode each frame of raw image data **201** as a respective frame of differential image data **202**. For example, the differential encoding component **212** may calculate a predicted value ($\{\tilde{x}\}$) associated with each pixel value (x) in a frame of raw image data **201** and may output only the difference (d) between the predicted value $\{\tilde{x}\}$ and the actual pixel value x (where $d = \{\tilde{x}\} - x$). More specifically, the differential encoding component **212** may assume that the raw image data **201** remains relatively unchanged across successive frames of video or follows a known pattern or statistical variation. As a result, the predicted values $\{\tilde{x}\}$ may be very close, if not identical, to the actual pixel values x (and thus, the differences d may be very small and often equal to zero).

(26) The data compression component **214** is configured to compress the differential image data **202** to further reduce the size of data transmissions from the image encoder **210** to the image decoder **220**. In some aspects, the data compression component **214** may encode each difference value d associated with the differential image data **202** as a variable-length codeword in accordance with an entropy encoding technique. In some implementations, the data compression component **214** may include an EGR encoder configured to output EGR codewords based, at least in part, on the differential image data **202**. As described above, the length of an EGR codeword depends on the size of the input value to be encoded, such that larger input values are encoded as longer EGR codewords and smaller input values are encoded as shorter EGR codewords. Because the difference values d are likely to be small, the data compression component **214** may achieve a relatively high compression ratio by performing EGR coding on the differential image data **202**.

(27) Aspects of the present disclosure recognize that the differential image data **202** often includes long runs of zeroes. As such, the compression ratio associated with the data compression component **214** can be further increased by encoding runs of zeroes in the differential image data **202** (such as with run-length encoding) rather than encoding each zero value individually. In some aspects, the data compression component **214** may be configured to encode each run of one or more zeroes in the differential image data **202** as an EGR codeword representing the number of consecutive zeroes and may encode each non-zero value in the differential image data **202** as an EGR codeword representing the non-zero value. For example, if the differential image data **202**

includes a run of 5 consecutive zeros, the data compression component **214** may encode the run of 5 zeroes as a single EGR codeword rather than 5 EGR codewords (each representing a value equal to zero).

(28) In some implementations, the data compression component **214** may add a zero-run (ZR) prefix to each EGR codeword representing a run of zeroes (also referred to herein as “zero-run codewords”) to differentiate such EGR codewords from other EGR codewords that represent non-zero values (also referred to herein as “non-zero codewords”). Aspects of the present disclosure recognize that all non-zero codewords include a unary prefix consisting of one or more bits having the same binary value. As such, the LSB of each non-zero codeword has the same value (either 0 or 1, depending on implementation). In some implementations, the ZR prefix may be set to a value that is different than the LSB of any non-zero codeword. For example, if each non-zero codeword has its LSB set to 0, the ZR prefix appended to each zero-run codeword is set to 1. Conversely, if each non-zero codeword has its LSB set to 1, the ZR prefix appended to each zero-run codeword is set to 0.

(29) In some implementations, the image decoder **220** may include a differential decoding component **222** and a data decompression component **224**. The data decompression component **224** is configured to decode the compressed image data **203** as decompressed image data **204**. More specifically, the data decompression component **224** may reverse the compression performed by the data compression component **214**. In some implementations, the data decompression component **224** may be configured to distinguish between zero-run codewords and non-zero codewords based, at least in part, on the presence or absence of a ZR prefix. For example, the data decompression component **224** may identify and decode non-zero codewords in the compressed image data **203** in accordance with existing EGR decoding techniques. However, if an EGR codeword is preceded by a ZR prefix, the data decompression component **224** may interpret the decoded value of that EGR codeword to indicate a number of consecutive zeroes.

(30) The differential decoding component **222** is configured to recover the reconstructed image data **205** from the decompressed image data **204**. More specifically, the differential decoding component **222** may reverse the encoding performed by the differential encoding component **212**. For example, the differential decoding component **222** may calculate a series of predicted values $\{\tilde{x}\}$ using the same prediction loop or algorithm as implemented by the differential encoding component **212**. In some implementations, the decompressed image data **204** may include a series of recovered difference values ($\{\hat{d}\}$) that are substantially similar, if not identical, to the difference values d associated with the differential image data **202**. Accordingly, the differential decoding component **222** may add the recovered difference values $\{\hat{d}\}$ to the series of predicted values $\{\tilde{x}\}$ to obtain a series of reconstructed values x , respectively, representing the reconstructed image data **205**.

(31) As described above, the data compression component **214** and the data decompression component **224** exploit an intrinsic property associated with certain types of entropy codes (such as those with a unary prefix) to signal whether a given codeword represents an integer value or a run of zeroes. In the example of FIG. 2, the encoding techniques are described specifically with respect to EGR codes. However, aspects of the present disclosure also may be implemented with respect to other encoding techniques. For example, a ZR prefix may be added to any codeword having a unary prefix (where the binary value of the ZR prefix is different than the binary value of the LSB of the codeword) to signal that the decoded value of the codeword represents a run of zeroes rather than a particular integer value.

(32) FIG. 3 shows a block diagram of a variable-length encoder **300**, according to some implementations. In some implementations, the variable-length encoder **300** may be one example of the data compression component **214** of FIG. 2. More specifically, the encoder **300** may be configured to receive a frame of data **302** and produce an encoded output **308** for transmission or storage via a channel (such as the channel **130** of FIG. 1).

(33) In some implementations, the variable-length encoder **300** may include an integer mapper **310**, a zero-run (ZR) detector **320**, and a zero-run compression (ZRC) encoder **330**. The integer mapper **310** is configured to map the data frame to a sequence of integer values **304** having a probability distribution concentrated around zero. For example, the data frame **302** may include a number (M) of data symbols having positive or negative values. In some aspects, the integer sequence **304** may have a probability distribution concentrated around zero. In other words, more commonly-occurring data symbols are mapped to smaller integer values and less commonly-occurring data symbols are mapped to larger integer values, with the most commonly-occurring data symbol being mapped to zero. In some implementations, the integer mapper **310** may map each of the M data symbols to a non-negative (or unsigned) integer value. In some other implementations, the integer mapper **310** may map each of the M data symbols to a signed integer value.

(34) The ZR detector **320** is configured to scan the integer sequence **304** for runs of consecutive zeroes. In some implementations, the ZR detector **320** may iteratively scan the integer sequence **304** (such as from the MSB to the LSB) so that a different integer value is scanned during each iteration. During each iteration of the scan, the ZR detector **320** may determine whether the current (scanned) integer value is equal to zero. If the current integer value is equal to zero, the ZR detector **320** may increment a counter **322** and proceed to scan the next integer value in the integer sequence **304**. As such, the counter **322** maintains a count value (N) indicating the number of consecutive zeroes scanned by the ZR detector **320**. If the current integer value is not equal to zero, the ZR detector **320** may provide an input value **306** to the ZRC encoder **330** based, at least in part, on the current value N associated with the counter **322**.

(35) In some implementations, the ZR detector **320** may output the current (non-zero) integer value in the integer sequence **304** or a value associated with the counter **322** as the input value **306**. More specifically, the ZR detector **320** may output the current integer value in the integer sequence **304** if the current count value N is equal to zero. For example, if the current integer value is equal to 5, the ZR detector **320** may output “5” as the input value **306**. On the other hand, if the current count value N is greater than zero, the ZR detector **320** may output a value equal to N-1 as the input value **306**. For example, if the current count value N is equal to 5, the ZR detector **320** may output “4” as the input value **306**. After outputting N-1 as the input value **306**, the ZR detector **320** may reset the counter **322** (to N=0) and subsequently output the current integer value in the integer sequence **304** (as the following input value **306**) before proceeding to the next iteration of the scan.

(36) In some aspects, the ZR detector **320** may further provide a control signal **307** to the ZRC encoder **330**. In some implementations, the control signal **307** may indicate whether the input value **306** represents a run of zeroes or a non-zero integer value. For example, the ZR detector **320** may drive the control signal **307** to a first state (indicating a non-zero integer value) when outputting the current integer value in the integer sequence **304** as the input value **306** and may drive the control signal **307** to a second state (indicating a run of zeroes) when outputting N-1 as the input value **306**. In some other implementations, the control signal **307** may indicate whether the ZR detector **320** has scanned all M integer values in the integer sequence **304**. For example, after completing the N^{sup}.th iteration of the scan, the ZR detector **320** may drive the control signal **307** to a third state (indicating that all M integer values in the integer sequence **304** have been scanned).

(37) The ZRC encoder **330** is configured to generate an encoded output based on the input value **306**, the control signal **307**, or a combination thereof. In some implementations, the ZRC encoder **330** may encode the input value **306** as a non-zero codeword responsive to the control signal **307** indicating that the input value **306** represents a non-zero integer value. For example, the ZRC encoder **330** may encode the input value **306** in accordance with an EGR code **332**. In some other implementations, the ZRC encoder **330** may encode the input value **306** as a zero-run codeword responsive to the control signal **307** indicating that the input value **306** represents a run of zeroes. For example, the ZRC encoder **330** may encode the input value **306** in accordance with the EGR code **332** and may further append a ZR prefix to the resulting EGR codeword (such as described

with reference to FIG. 2).

(38) Still further, in some implementations, the ZRC encoder **330** may output a stop code (as the encoded output **308**) responsive to the control signal **307** indicating that all M integer values in the integer sequence **304** have been scanned. For example, the stop code may include a unique sequence of bits that cannot be decoded as a non-zero codeword or as a zero-run codeword. In other words, the stop code may be different than any non-zero codeword or zero-run codeword that can be output by the ZRC encoder **330**. In some implementations, the stop code may consist of three consecutive bits each having a binary value equal to 1 (such as “111”).

(39) FIG. 4 shows an example zero-run compression (ZRC) encoder **400**, according to some implementations. In some implementations, the ZRC encoder **400** may be one example of the ZRC encoder **330** of FIG. 3. More specifically, the ZRC encoder **400** represents a logical implementation of the ZRC encoder **330** of FIG. 3. In the example of FIG. 4, the ZRC encoder **400** is shown to include an entropy encoder **410**, a ZR prefix adder **420**, and an output selector (or multiplexer) **430**.

(40) The entropy encoder **410** is configured to receive the input value **306** from the ZR detector **320** of FIG. 3 and output a non-zero codeword **402** based on the input value **306**. In some implementations, the entropy encoder **410** may encode the input value **306** in accordance with an entropy encoding scheme. For example, the entropy encoding scheme maybe an EGR code. As such, the non-zero codeword **402** may be an EGR codeword having a unary prefix consisting of 0's (or 1's). The ZR prefix adder **420** is configured to append a ZR prefix **401** to the encoded integer value **402** to produce a zero-run codeword **404**. More specifically, the ZR prefix **401** may represent the LSB(s) of the zero-run codeword **404**. In some implementations, the ZR prefix **401** may include one or more bits having a binary value that is different than the binary value associated with each bit of the unary prefix. For example, if each bit of the unary prefix is set to 0, the ZR prefix **401** may include one or more bits that are each set to 1.

(41) The output selector **430** is configured to select one of the non-zero codeword **402**, the zero-run codeword **404**, or a stop code **406** for output as the encoded output **308**. As described with reference to FIG. 3, the stop code **406** may be a string of bits that cannot be interpreted as a non-zero codeword **402** or a zero-run codeword **404**. In some implementations, the output selector **430** may select the encoded output **308** based on the state of the control signal **307**. For example, the output selector **430** may output the non-zero codeword **402** if the control signal **307** is driven to a first state (indicating that the input value **306** represents a non-zero integer value), may output the zero-run codeword **404** if the control signal **307** is driven to a second state (indicating that the input value **306** represents a run of zeroes), and may output the stop code **406** if the control signal **307** is driven to a third state (indicating that all M integer values in the integer sequence **304** have been scanned).

(42) FIG. 5 shows an example process **500** for encoding data, according to some implementations. In some implementations, the encoding process **500** may be performed by the variable-length encoder **300** of FIG. 3. For example, the encoder **300** may receive a data frame **502** and may produce an encoded output **506** based on the received data frame.

(43) In the example of FIG. 5, the data frame **502** is shown to include a sequence of 24 data symbols “ABCAAAAADDDCAAAAAAAAAAAAA.” In some implementations, each data symbol may represent a pixel value or a pixel differential (representing a difference between a pixel value and a reference signal) associated with a respective pixel of an image or video frame. With reference for example to FIG. 3, the integer mapper **310** may map the 24 data symbols to 24 integer values, respectively, in an integer sequence **504**. As shown in FIG. 5, the values associated with the integer sequence **504** follow a normal probability distribution concentrated around zero. That is, the integer value 0 has the highest probability of occurring, the integer value 1 has the second-highest probability of occurring, the integer value 2 has the third-highest probability of occurring, and this trend may continue on for any range of integer values.

(44) Accordingly, the integer mapper **310** maps the most commonly-occurring value (“A”)

associated with the data frame **502** to the integer value 0, maps the second-most commonly-occurring data symbol (“D”) associated with the data frame **502** to the integer value 1, maps the third-most commonly-occurring data symbol (“C”) associated with the data frame **502** to the integer value 2, and maps the fourth-most commonly-occurring data symbol (“B”) associated with the data frame **502** to the integer value 3. As a result of the mapping, the integer sequence **504** includes a sequence of 24 integer values “032000001112000000000000.”

(45) With reference for example to FIG. **3**, the ZR detector **320** may iteratively scan the integer sequence **504** (such as from MSB to LSB) for runs of zeroes. During the 1.sup.st iteration of the scan, the ZR detector **320** detects that the current integer value is 0 to zero and, in response thereto, increments the counter **320** (so that N=1). During the 2.sup.nd iteration of the scan, the ZR detector **320** detects that the current integer value is equal to 3 and, in response thereto, outputs an input value **306** equal to 0 (representing N-1) while driving the control signal **307** to the second state (indicating a run of zeroes). Thereafter, the ZR detector **320** may reset the counter **322** (to N=0). The ZRC encoder **330** encodes the input value **306** as the EGR codeword “1” and appends a ZR prefix (“1”) to the EGR codeword responsive to the control signal **307** being driven to the second state. As a result, “11” is the first codeword associated with the encoded output **506**.

(46) The ZR detector **320** further outputs an input value **306** equal to 3 (representing the current non-zero integer value) while driving the control signal **307** to the first state (indicating a non-zero integer value). The ZRC encoder **330** encodes the input value **306** as the EGR codeword “1100” but does not add the ZR prefix since the control signal **307** is in the first state. As a result, “1100” is the second codeword associated with the encoded output **506**. During the 3rd iteration of the scan, the ZR detector **320** detects that the current integer value is equal to 2. Because the current count value N is equal to zero, the ZR detector **320** outputs an integer value **306** equal to 2 (representing the current non-zero integer value) while driving the control signal **307** to the first state. The ZRC encoder **330** encodes the input value **306** as the EGR codeword “0010” but does not add the ZR prefix since the control signal **307** is in the first state. As a result, “0010” is the third codeword associated with the encoded output **506**.

(47) During each of the 4th 5th 6th 7th and 8th iterations of the scan, the ZR detector **320** detects that the current integer value is 0 to zero and, in response thereto, increments the counter **320** (so that N=5 by the end of the 8th iteration). During the 9th iteration of the scan, the ZR detector **320** detects that the current integer value is equal to 1 and, in response thereto, outputs an input value **306** equal to 4 (representing N-1) while driving the control signal **307** to the second state. Thereafter, the ZR detector **320** may reset the counter **322** (to N=0). The ZRC encoder **330** encodes the input value **306** as the EGR codeword “001000” and appends the ZR prefix (“1”) to the EGR codeword responsive to the control signal **307** being driven to the second state. As a result, “0010001” is the fourth codeword associated with the encoded output **506**.

(48) The ZR detector **320** further outputs an input value **306** equal to 1 (representing the current non-zero integer value) while driving the control signal **307** to the first state. The ZRC encoder **330** encodes the input value **306** as the EGR codeword “10” but does not add the ZR prefix since the control signal **307** is in the first state. As a result, “10” is the fifth codeword associated with the encoded output **506**. During the 10th iteration of the scan, the ZR detector **320** detects that the current integer value is equal to 1. Because the current count value N is equal to zero, the ZR detector **320** outputs an integer value **306** equal to 1 (representing the current non-zero integer value) while driving the control signal **307** to the first state. The ZRC encoder **330** encodes the input value **306** as the EGR codeword “10” but does not add the ZR prefix since the control signal **307** is in the first state. As a result, “10” is the sixth codeword associated with the encoded output **506**.

(49) During the 11th iteration of the scan, the ZR detector **320** detects that the current integer value is equal to 1. Because the current count value N is still equal to zero, the ZR detector **320** outputs an integer value **306** equal to 1 while driving the control signal **307** to the first state. The ZRC

encoder **330** encodes the input value **306** as the EGR codeword “10” but does not add the ZR prefix since the control signal **307** is in the first state. As a result, “10” is the seventh codeword associated with the encoded output **506**. During the 12th iteration of the scan, the ZR detector **320** detects that the current integer value is equal to 2. Because the current count value N is still equal to zero, the ZR detector **320** outputs an integer value **306** equal to 2 while driving the control signal **307** to the first state. The ZRC encoder **330** encodes the input value **306** as the EGR codeword “0010” but does not add the ZR prefix since the control signal **307** is in the first state. As a result, “0010” is the eighth codeword associated with the encoded output **506**.

(50) During each of the 13.sup.th through 24.sup.th iterations of the scan, the ZR detector **320** detects that the current integer value is 0 to zero and, in response thereto, increments the counter **320** (so that N=12 by the end of the 24.sup.th iteration). However, because all 24 integer values associated with the integer sequence **504** have been scanned upon completion of the 24.sup.th iteration of the scan, the ZR detector **320** does not output a subsequent input value **306**. Rather, the ZR detector **320** may drive the control signal **307** to the third state (indicating that all integer values in the integer sequence have been scanned). The ZRC encoder **330** outputs the stop code “111” responsive to the control signal **307** being driven to the third state. As a result, “111” is the ninth and final codeword associated with the encoded output **506**.

(51) FIG. **6** shows a block diagram of an encoder **600**, according to some implementations. In some implementations, the encoder **600** may be one example of any of the encoders **110**, **210**, or **300** of FIGS. **1**, **2**, and **3**, respectively. More specifically, the encoder **600** may be configured to encode data symbols using entropy encoding techniques with run-length encoding of zeroes.

(52) In some implementations, the encoder **600** may include a data interface **610**, a processing system **620**, and a memory **630**. The data interface **610** is configured to receive the data symbols and output encoded codewords associated with the data symbols. In some aspects, the data interface **610** may include a data source interface (I/F) **612** and a channel interface **614**. The data source interface **612** is configured to communicate with a data source coupled to or otherwise associated with the encoder **600**. In some implementations, the data source interface **612** may obtain a number (M) of integer values representing a frame of data. The channel interface **614** is configured to communicate over a channel coupled between the encoder **600** and a decoder (not shown for simplicity).

(53) The memory **630** may include a data buffer **632** configured to store the received data symbols and any intermediate data associated with the encoding operation. The memory **630** also may include a non-transitory computer-readable medium (including one or more nonvolatile memory elements, such as EPROM, EEPROM, Flash memory, a hard drive, and the like) that may store at least the following software (SW) modules: a zero detection SW module **634** to perform a zero-detection operation that scans the M integer values over M iterations, respectively, and indicate, during each of the M iterations, whether the scanned integer value is equal to zero; a ZR counter SW module **636** to increment a count value (N) responsive to the zero-detection operation indicating that the scanned integer value is equal to zero; and a variable-length encoding SW module **638** to output a first pattern of bits responsive to the zero-detection operation indicating that the scanned integer value is not equal to zero, where the first pattern of bits represents one of the count value N or the scanned integer value based on whether N is greater than zero. Each software module includes instructions that, when executed by the processing system **620**, causes the encoder **600** to perform the corresponding functions.

(54) The processing system **620** may include any suitable one or more processors capable of executing scripts or instructions of one or more software programs stored in the encoder **600** (such as in memory **630**). For example, the processing system **620** may execute the zero detection SW module **634** to perform a zero-detection operation that scans the M integer values over M iterations, respectively, and indicate, during each of the M iterations, whether the scanned integer value is equal to zero. The processing system **620** also may execute the ZR counter SW module **636** to

increment a count value (N) responsive to the zero-detection operation indicating that the scanned integer value is equal to zero. Further, the processing system **620** may execute the variable-length encoding SW module **638** to output a first pattern of bits responsive to the zero-detection operation indicating that the scanned integer value is not equal to zero, where the first pattern of bits represents one of the count value N or the scanned integer value based on whether N is greater than zero.

(55) FIG. 7 shows an illustrative flowchart depicting an example operation **700** for data encoding, according to some implementations. In some implementations, the example operation **700** may be performed by an encoder such as any of the encoders **110**, **210**, **300** or **600** of FIGS. 1, 2, 3, and 6 respectively.

(56) The encoder may obtain a number (M) of integer values representing a frame of data (**710**). The encoder may perform a zero-detection operation that scans the M integer values over M iterations, respectively, and indicates, during each of the M iterations, whether the scanned integer values is equal to zero (**720**). The encoder also may increment a count value (N) responsive to the zero-detection operation indicating that the scanned integer value is equal to zero (**730**). The encoder may further output a first pattern of bits responsive to the zero-detection operation indicating that the scanned integer value in the sequence is not equal to zero, where the first pattern of bits represents one of the count value N or the scanned integer value based on whether N is greater than zero (**740**).

(57) In some aspects, the obtaining of the M integer values may include receiving a sequence of data symbols and mapping each data symbol in the sequence to a respective integer value of the M integer values. In some implementations, each of the M integer values may be a non-negative value.

(58) In some aspects, the outputting of the first pattern of bits may include determining that N is not greater than zero and encoding the scanned integer value as the first pattern of bits responsive to determining that the count value N is not greater than zero. In some implementations, the first pattern of bits may be an exponential Golomb-Rice codeword representing the scanned integer value.

(59) In some aspects, the outputting of the first pattern of bits may include determining that N is greater than zero and encoding N-1 as an exponential Golomb-Rice codeword responsive to determining that the count value is greater than zero. In some implementations, the first pattern of bits may include a zero-run prefix followed by the exponential Golomb-Rice codeword. In some implementations, the zero-run prefix may be a single bit having a different binary value than the LSB of the first pattern of bits. In some implementations, the encoder may further encode the scanned integer value as a second pattern of bits, output the second pattern of bits following the first pattern of bits, and reset the count value to N=0.

(60) In some aspects, the encoder may further output a stop code responsive to completing the M.sup.th iteration of the zero-detection operation. In some implementations, the stop code may include three consecutive bits each having a different binary value than the LSB of the first pattern of bits.

(61) Those of skill in the art will appreciate that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(62) Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their

functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

(63) The methods, sequences or algorithms described in connection with the aspects disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor.

(64) In the foregoing specification, embodiments have been described with reference to specific examples thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader scope of the disclosure as set forth in the appended claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A method of data encoding, comprising: obtaining a number (M) of integer values representing a frame of data; performing a zero-detection operation that scans the M integer values over M iterations, respectively, and indicates, during each of the M iterations, whether the scanned integer value is equal to zero; incrementing a count value (N) responsive to the zero-detection operation indicating that the scanned integer value is equal to zero; determining whether the count value N is greater than zero responsive to the zero-detection operation indicating that the scanned integer value is not equal to zero; outputting a first pattern of bits, including an exponential Golomb-Rice (EGR) codeword, based on whether the count value N is determined to be greater than zero, the outputting of the first pattern of bits including: determining a zero-run prefix based at least in part on the least significant bit (LSB) of the first pattern of bits so that the zero-run prefix has a different binary value than the LSB of the pattern bits, the zero-run prefix indicating whether the EGR codeword represents a number of consecutive zeroes; encoding the scanned integer value as the EGR codeword and refraining from including the zero-run prefix in the first pattern of bits responsive to determining that the count value N is not greater than zero; and encoding N-1 as the EGR codeword and including the zero-run prefix in the first pattern of bits responsive to determining that the count value N is greater than zero; and outputting one or more consecutive bits representing a stop code responsive to completing the Mth iteration of the zero-detection operation, each of the one or more consecutive bits each having a different binary value than the LSB of the first pattern of bits.

2. The method of claim 1, wherein the obtaining of the M integer values comprises: receiving a sequence of data symbols; and mapping each data symbol in the sequence to a respective integer value of the M integer values.

3. The method of claim 2, wherein each of the M integer values is a non-negative value.

4. The method of claim 1, wherein the zero-run prefix is followed by the EGR codeword in the first pattern of bits.

5. The method of claim 4, wherein the zero-run prefix is a single bit.

6. The method of claim 1, further comprising: encoding the scanned integer value as a second pattern of bits; outputting the second pattern of bits following the first pattern of bits; and resetting the count value to N=0.

7. The method of claim 1, wherein the stop code comprises three consecutive bits.

8. An encoder comprising: a processing system; and a memory storing instructions that, when executed by the processing system, causes the encoder to: obtain a number (M) of integer values representing a frame of data; perform a zero-detection operation that scans the M integer values over M iterations, respectively, and indicates, during each of the M iterations, whether the scanned integer value is equal to zero; increment a count value (N) responsive to the zero-detection operation indicating that the scanned integer value is equal to zero; determine whether the count value N is greater than zero responsive to the zero-detection operation indicating that the scanned integer value is not equal to zero; and output a first pattern of bits, including an exponential Golomb-Rice (EGR) codeword, based on whether N is greater than zero, the outputting of the first pattern of bits including: determining a zero-run prefix based at least in part on the least significant bit (LSB) of the first pattern of bits so that the zero-run prefix has a different binary value than the LSB of the first pattern of bits, the zero-run prefix indicating whether the EGR codeword represents a number of consecutive zeroes; encoding the scanned integer value as the EGR codeword and refraining from including the zero-run prefix in the first pattern of bits responsive to determining that the count value N is not greater than zero; and encoding N-1 as the EGR codeword and including the zero-run prefix in the first pattern of bits responsive to determining that the count value is greater than zero; and output one or more consecutive bits representing a stop code responsive to completing the Mth iteration of the zero-detection operation, each of the one or more consecutive bits each having a different binary value than the LSB of the first pattern of bits.
9. The encoder of claim 8, wherein the obtaining of the M integer values comprises: receiving a sequence of data symbols; and mapping each data symbol in the sequence to a respective integer value of the M integer values.
10. The encoder of claim 8, wherein the zero-run prefix is a single bit.
11. The encoder of claim 8, wherein execution of the instructions further causes the encoder to: encode the scanned integer value as a second pattern of bits; output the second pattern of bits following the first pattern of bits; and reset the count value to N=0.
12. The encoder of claim 8, wherein the stop code comprises three consecutive bits.
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